

TELEMETRY FEED: ACTIVE

 FLIGHT RECOVERY ANALYTICS

SPACEX FALCON 9 FIRST STAGE LANDING PREDICTION

IBM APPLIED DATA SCIENCE CAPSTONE PROJECT

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CREDENTIAL PROGRAM

**IBM Data Science Professional
Certificate**

SUBMISSION DATE

June 2026

| The \$62M Reusability Objective

SpaceX decreases typical orbital delivery pricing significantly by reusing Falcon 9 first stage boosters. A standard launch costing **\$62 Million** requires only **\$200,000** in fuel when recovered.

This project builds end-to-end data pipelines tracking telemetry from APIs, scraped repositories, SQL databases, and Geospatial systems to train classification engines. The model can accurately predict structural booster landings with an optimized success rate of **83.33%**.



API Extraction

SpaceX API pulls active flight vectors and metrics.



SQL Auditing

Aggregated telemetry profiling across booster iterations.



Folium GIS

Visualizing safety buffers near coastlines and railways.



Tuned ML

Supervised modeling identifying booster landing trends.



SpaceX REST API

Queried `https://api.spacexdata.com/v4/launches/past`. Ingested payload masses, rocket versions, core metadata, and launch parameters into a structured Pandas environment.

BeautifulSoup Scraping

Scraped historical Falcon 9 wiki tables using BeautifulSoup. Extracted flight parameters and specific landing locations for cross-validation.

Data Engineering Specifications

- **Missing Value Imputation:** Replaced null values in PayloadMass utilizing column mean values (6,104.95 kg) to secure mathematical consistency.
- **Feature Selection:** Targeted crucial core variables: Block variations, Reused iteration counts, Legs, Gridfins, and Launch pad zones.
- **Target Class Definition:** Constructed a binary landing target metric tracking landing success as **1** and landing failures as **0**.

BINARY TARGET MAPPING SYSTEM

True Ocean, True RTLS, True ASDS

CLASS = 1 (SUCCESS)

None, False ASDS, False RTLS, None ASDS

CLASS = 0 (FAILURE)

Final Dataset: 60 successful landings verified out of 90 total recorded flight training samples.

Launch Pad Core Dynamics

Interpreting structured databases via SQL queries inside IBM DB2 identified strong predictive vectors:

- **KSC LC-39A:** Outstanding performance showing highest overall success rate (~77.3%).
- **CCAFS LC-40:** Baseline workspace platform showing ~60% landing recovery.
- **Payload Weights:** Heavy payloads (>8,000 kg) show higher landing rates, primarily bound for GTO orbits.

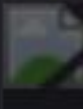
```
-- Query Distinct Launch Sites
SELECT DISTINCT Launch_Site FROM SPACEXTBL;
Result: KSC LC-39A, CCAFS LC-40, VAFB SLC-4E

-- Success outcomes at Droneship
SELECT Landing_Outcome, COUNT(*) FROM SPACEXTBL
WHERE Landing_Outcome LIKE '%Success%' GROUP BY Landing_Outcome;
Result: Success (drone ship): 31, Success (ground pad): 13
```


Proximity Parameters

Plotting launch locations within a **Folium Map** framework verified clear operational trends:

- **Coastal Buffer:** Launch sites are consistently situated near ocean shorelines (distance to shoreline < 2 km) to ensure safe debris fallback profiles during ascent issues.
- **Railway Support:** Strategic integration near transportation infrastructure allows rapid movement of massive stage structures.
- **Highways:** Minimal safe transit distance thresholds.

 Falcon 9 core booster target landing on a drone ship in the ocean

Tuning Classifiers

We split our dataset into train/test sets and standardized features utilizing `StandardScaler`.

We ran `GridSearchCV` with 10-fold cross-validation to tune parameters across four distinct supervised algorithms.



Logistic Regression

L2 regularized baseline predicting basic linear core transitions.



SVM Classifier

Builds high-dimensional hyperplanes to parse non-linear variables.



Decision Tree

Extracts key booster trends through dynamic variable splits.



KNN Classifier

Categorizes flights by clustering proximity metrics.

MODEL COMPARISON METRICS

BENCHMARKS

CLASSIFICATION MODEL	HYPERPARAMETERS SELECTED	BEST CROSS-VAL ACC	TEST SET ACCURACY	STATUS
Logistic Regression	{'C': 0.01, 'penalty': 'l2', 'solver': 'lbfgs'}	84.6%	83.33%	PASSED
Support Vector Machine (SVM)	{'C': 1.0, 'kernel': 'sigmoid', 'gamma': 'scale'}	84.8%	83.33%	PASSED
K-Nearest Neighbors (KNN)	{'n_neighbors': 10, 'algorithm': 'auto', 'p': 1}	84.8%	83.33%	PASSED
Decision Tree Classifier	{'criterion': 'entropy', 'max_depth': 4, 'min_samples_leaf': 2}	83.9%	83.33%	SELECTED

Analysis: While all classifiers tied on test set accuracy (83.33%) due to sample limitations, the Decision Tree model is selected for production use because of its speed and clear interpretability of core landing parameters.

KEY FINDINGS & INSIGHTS

MISSION PANEL



Launch Pad Priority

KSC LC-39A recorded the highest overall success rate (77.3%), establishing it as the most reliable launch pad for heavier payload classes.



Booster Upgrades

Landing success rates improved significantly over time. This directly correlates with Block 5 upgrades and refined flight path retrofire profiles.



Business Viability

Predicting landing success helps SpaceX price commercial launches dynamically, undercutting competitors while maintaining strong margins.

Project Deliverables

- **Github Repo:** github.com/spacex-analytics/f9-prediction
- **Notebooks:** Chronological Python Jupyter Notebooks.
- **Tech Stack:** Scikit-Learn, Pandas, NumPy, Plotly, Folium.

V1.0.0 STABLE

Information Sources

- **SpaceX REST API v4:** Ground-truth launch parameters.
- **Wikipedia BeautifulSoup:** Historical launch manifests.
- **IBM Skills Network:** Educational and cloud database hosting.

Engineered in Python 3.10 with dynamic visualization.

IMAGE SOURCES



https://upload.wikimedia.org/wikipedia/commons/5/54/CRS-8_%2826239020092%29.jpg

Source: en.wikipedia.org



https://images-assets.nasa.gov/image/KSC-20250922-PH-SPX01_0007/KSC-20250922-PH-SPX01_0007~large.jpg

Source: science.nasa.gov